Catpus

Campus Cat Support Service

Team 2

박진석 박보연 박지수 오정호 조현우
최민성

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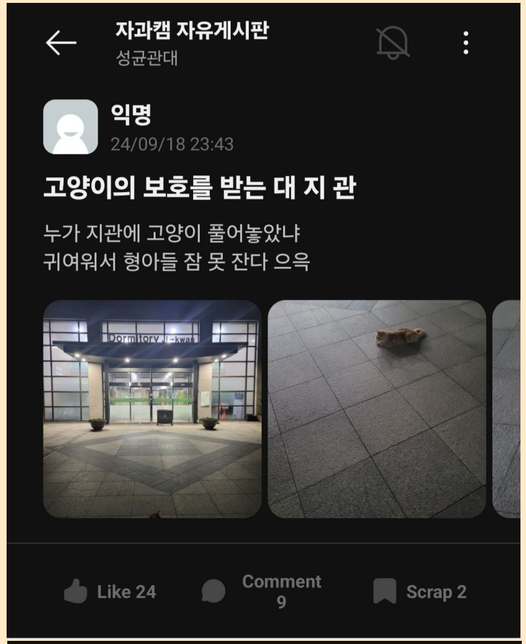
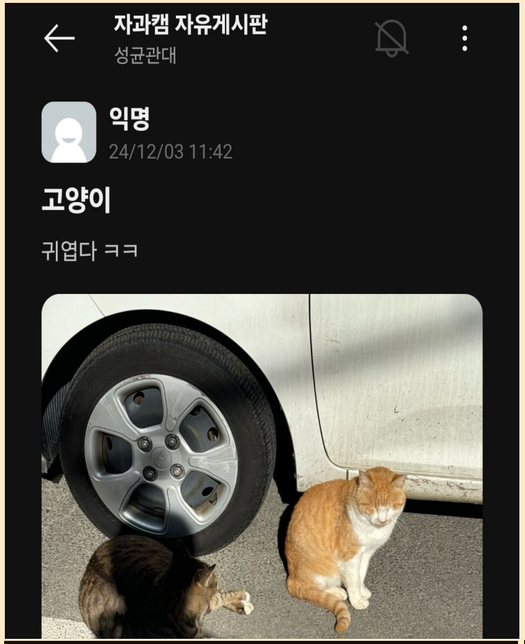
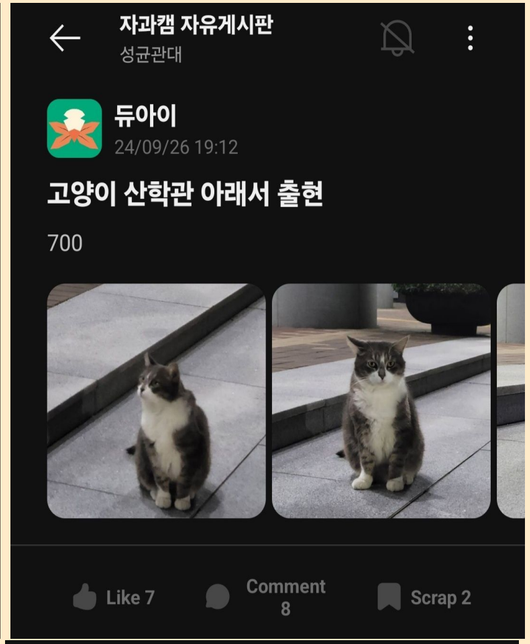
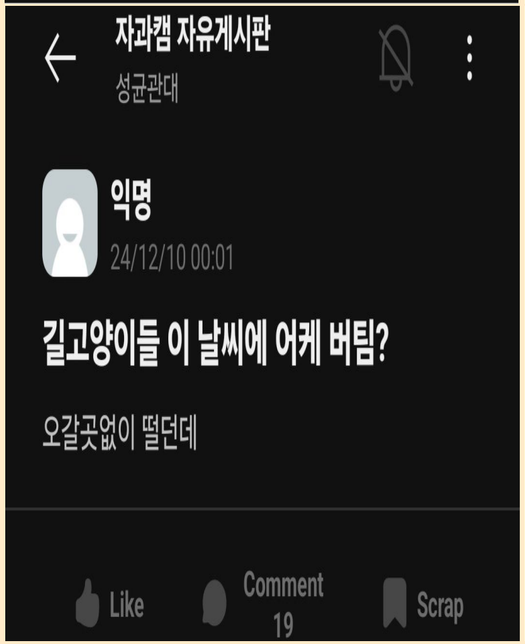
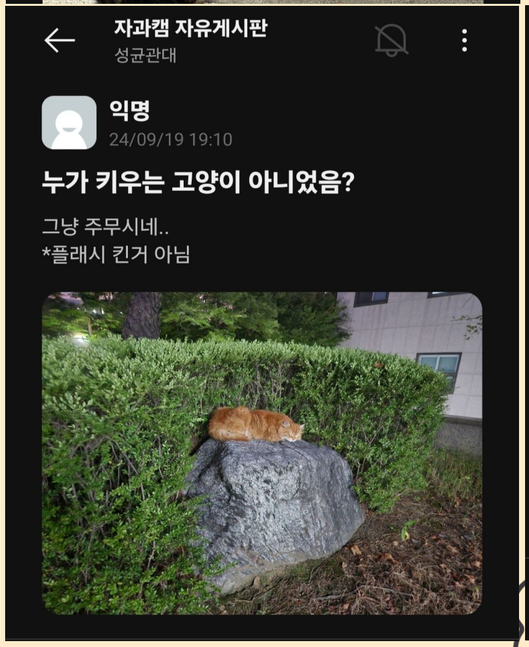
1. Project Motivation

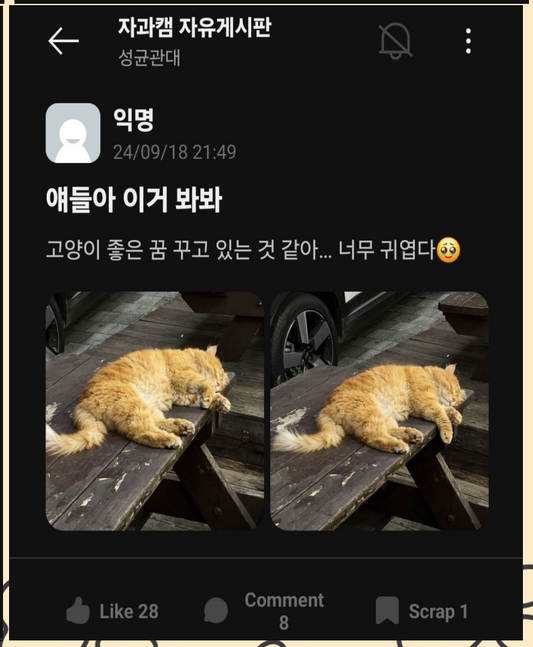












Background

Campus Cat's receive a lot of attention!

But are we truly living in harmony with them on our campus?

🐱🤔 ?

Project Motivation

Problem Recognition

- 🧍 **Loneliness** and **Stress** among University Students
- 🚫 **No Pets Allowed** in Dormitories
- 🐾 Strong Desire for **Animal Interaction**

Limitations of Existing Efforts

- ⏳ rely on **volunteer-based efforts**, lacking sustainability and trust
- 💸 **Commercial pet-sitting services** are costly

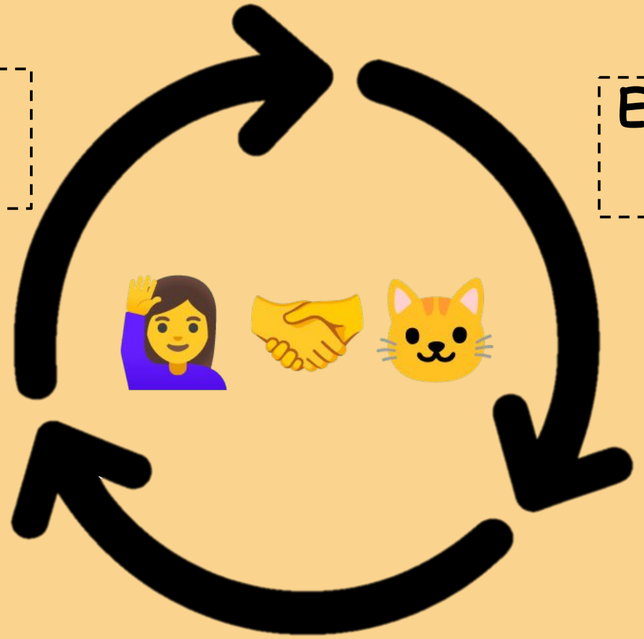
Aim to build a community where <u>students and animals</u> live in harmony

2. Project Purpose

Project Purpose

Build a sustainable care community where students and animals coexist

Establish a trusted system for pet sitting and volunteering



Foster empathy and responsibility through animal interaction on campus

3.Implementation Result

Member Roles

Backend

- 박진석
- 박보연

Frontend

- 오정호
- 조현우
- 최민성

Design

- 박지수

Technology Stack

Frontend (Mobile App)
- Framework: Flutter
- Language: Dart
- Supported OS: Android, iOS
- Mapping API: Google Maps SDK / OpenStreetMap
- Push Notification: Firebase Cloud Messaging (FCM),
  Apple Push Notification Service (APNs)

Backend (Server)
- Language & Framework: Node.js + Express.js
- Authentication: JWT (JSON Web Token)
- Real-time: WebSocket (Socket.io or native WebSocket)
- CI/CD: GitHub Actions (Lint, Unit Test, Integration Test)
- Containerization: Docker, Docker Compose

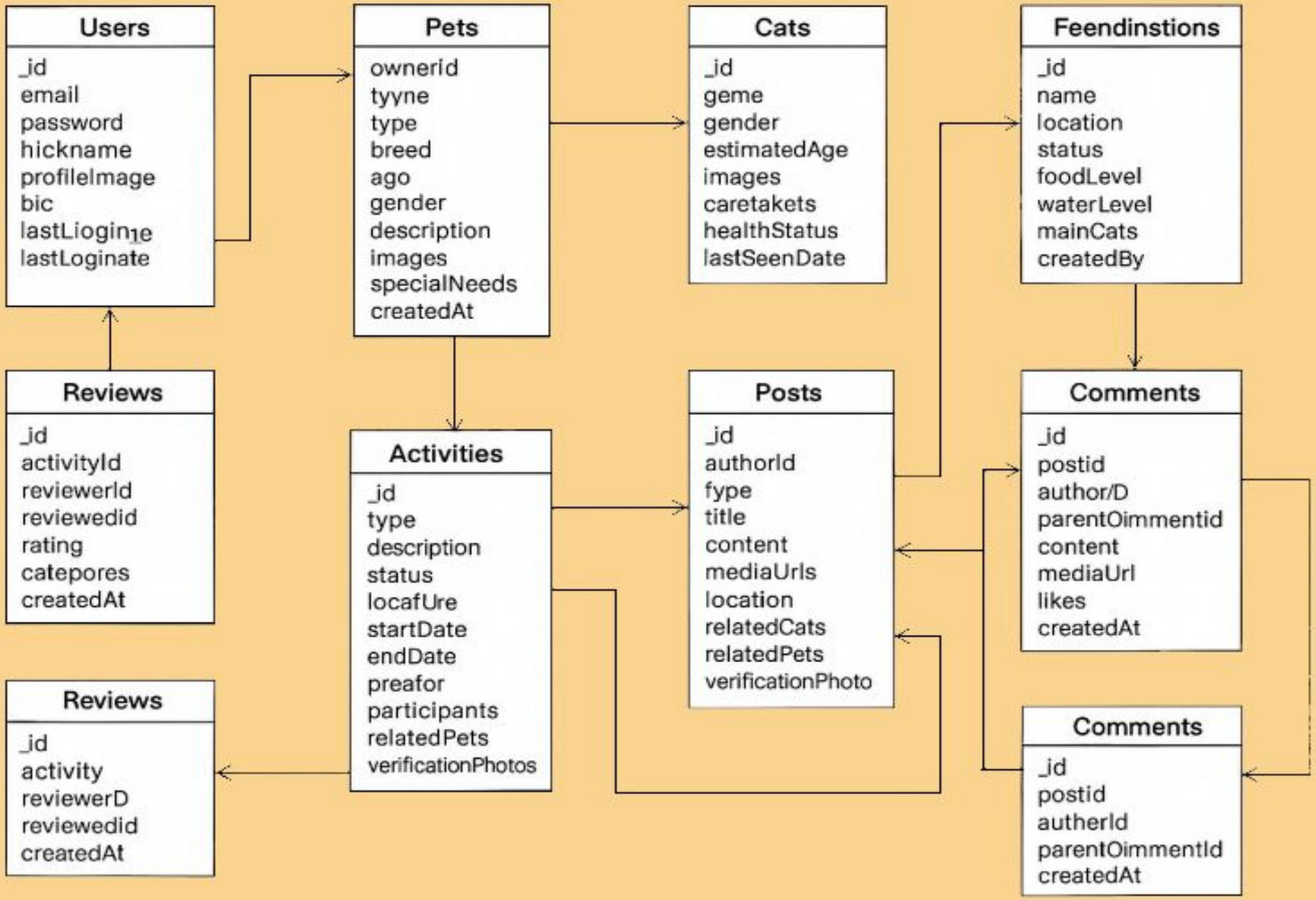
Database
- Primary DB: MongoDB
  - Configured as a 3-node replica set to ensure high availability
  - Example collections: Users, Pets, FeedingStations,
    Activities, Reviews, Posts, Comments, Badges,

Notifications
- Cache / Autocomplete: Redis
- Search Engine: Elasticsearch

External Services
- Cloud Storage: AWS S3
- Email Service: Amazon SES
- Geocoding / Reverse Geocoding: NCP Geocode API
- CDN: AWS CloudFront
- Monitoring & Logging: Kibana (Elastic Stack), AWS CloudWatch

Security
- Encryption: TLS 1.3 (communication encryption),
  AES-256 (data encryption at-rest)
- Vulnerability Management: periodic security audits based on
  OWASP standards
- Role-Based Access Control : separation of privileges between
  Administrator (System Manager)
  General User (General User)

ER diagram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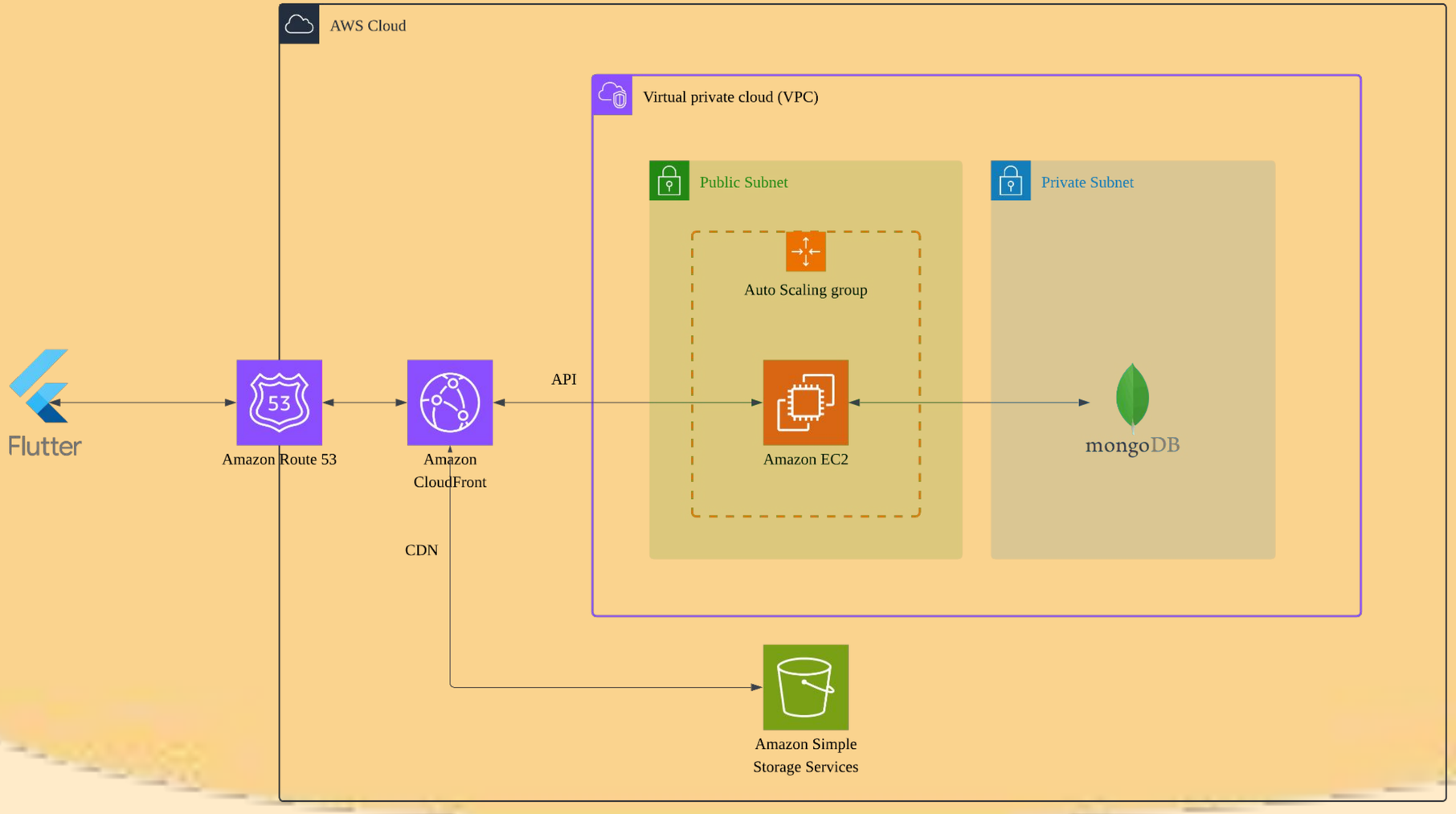
System Architecture



Application Features

0. Create Account & Settings
1. Feeding Campus Cat
2. Matching Pet Sitter
3. Looking for Volunteer Work
4. My Petstagram

0. Create Account & Settings

Start using Catpus with Easy Sign Up & Login !!

We support secure logins with email or social accounts

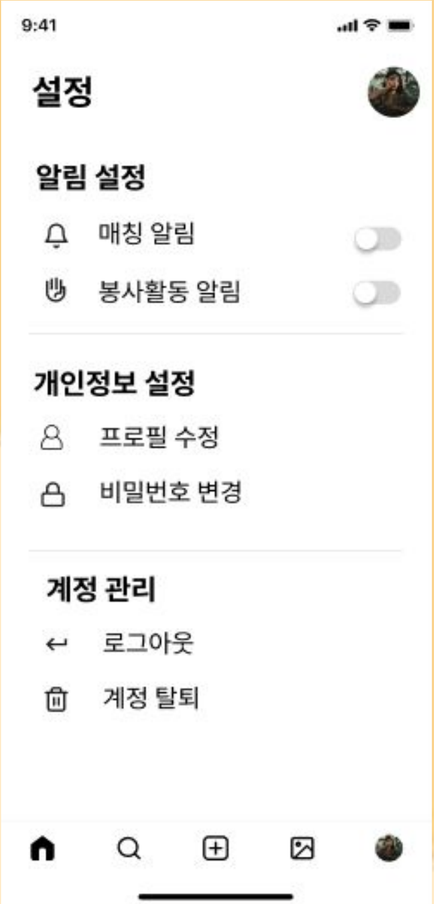
✔️ Email Verification – Authenticate with your campus email

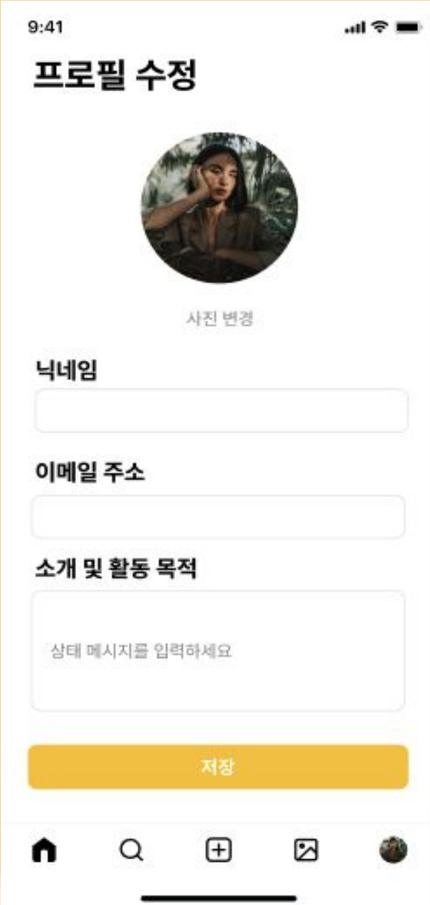
✔️ Quick Registration – Set a nickname, choose your campus and activity

✔️ Google/Apple Login – Sign up in seconds with your existing accounts

Join your campus pet community effortlessly and securely!

0. Create Account & Settings





Easily manage your profile and account !!

✔ Profile Management - Edit your profile, nickname ..

✔ Password & Account Control - Update your password, log out securely.

✔ Notification Settings - Turn on alerts for pet sitting matches or volunteer opportunities

Stay in control of your experience with just a few taps.

1. Feeding Campus Cat

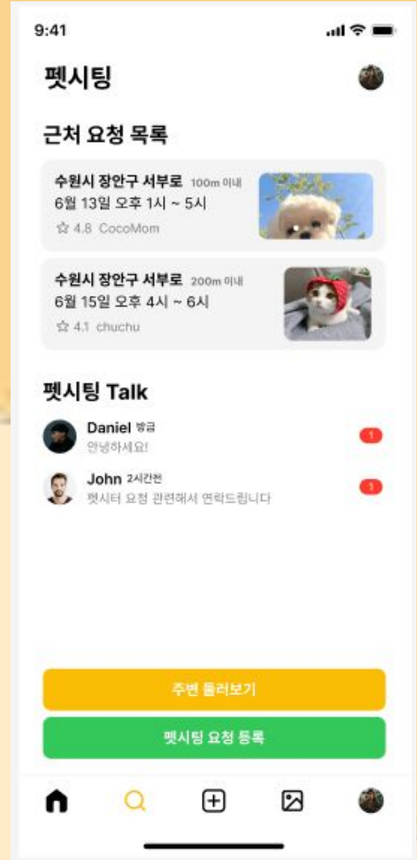
Help take care of stray cats around your campus !!

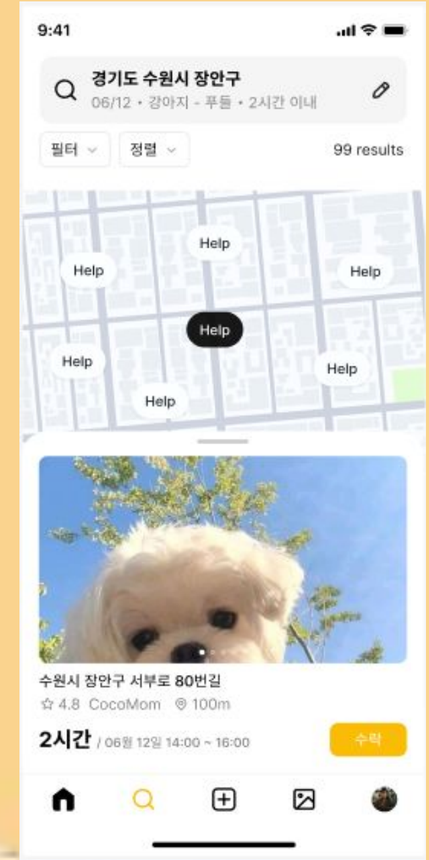
Check nearby feeding stations and contribute by feeding local cats.

- ✔ View exact feeding locations on the map
- ✔ See detailed directions and site photos

Make your campus a Warmer Place!

2. Matching Pet Sitter





Find the perfect Pet Sitter near you !!

Easily browse nearby pet sitting requests or offer your help as a sitter.

- ✔ Filter by pet type, time, and distance
- ✔ View sitter profiles and ratings
- ✔ Chat directly to coordinate details

Make pet care safer, faster, and more convenient — for both pet owners and sitters.

3. Looking for Volunteer Work

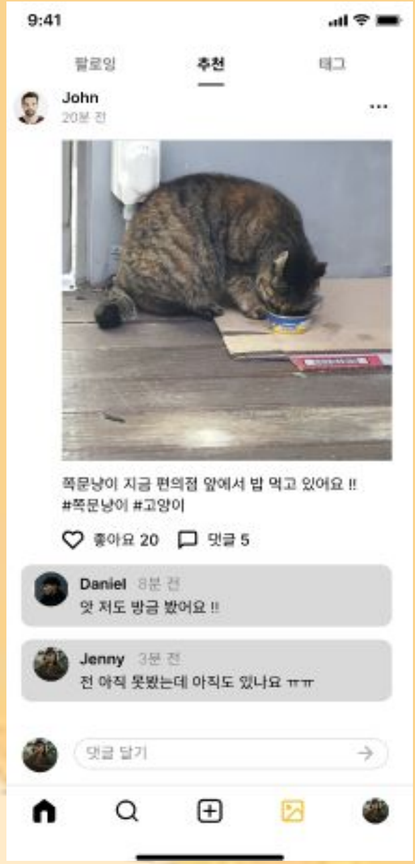
Discover and apply for Local Volunteer Activities !!

Browse available programs, check details, and submit your application — all in one place.

- ✔️ Search by date and location
- ✔️ View detailed descriptions and directions
- ✔️ Apply with a simple introduction

Make meaningful contributions to your community, one click at a time.

4. My Petstagram



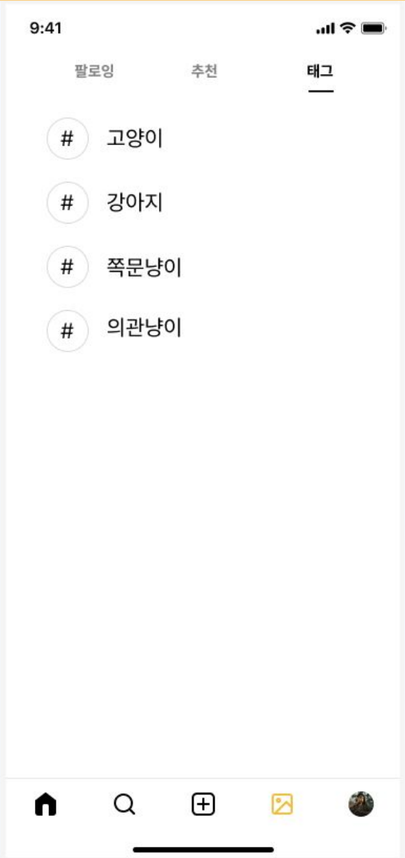
Share the daily life of your pets !!

Post cute photos, follow pet owners, explore popular hashtags, and leave comments.

- ✔️ Upload and browse pet posts
- ✔️ Follow and interact with other users
- ✔️ Discover trending tags like #puppy, #cat, and more

Turn your pet's moments into memories, and build your own pet-loving community.

4. My Petstagram - HashTag



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4. Persona & Demo Video

Persona

Cha Eun-woo (차은우)

Age: 20

Major: Computer Science & Engineering

Freshman from Jeju

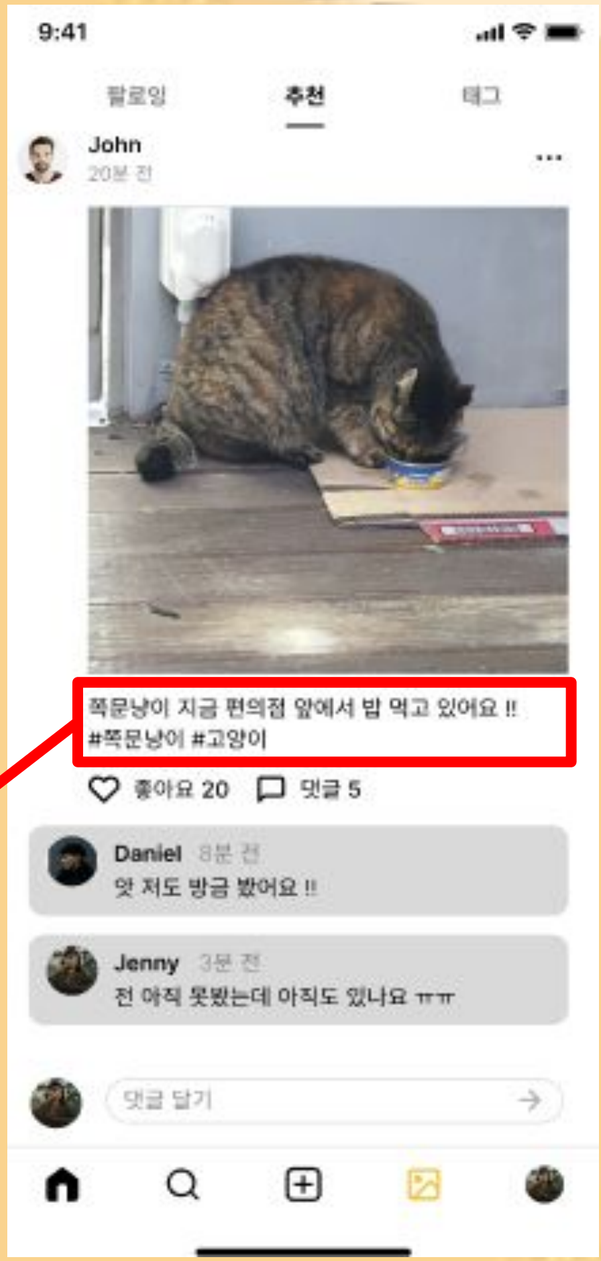
1. Loves cats & cares for 2 at home
2. First time living alone, homesick, misses cats
3. Strong desire for animal interaction…

"I want an easy way to connect with cats around campus!"

Scenario 1.

"I'm feeling lonely… I wonder where the brown cat is?"

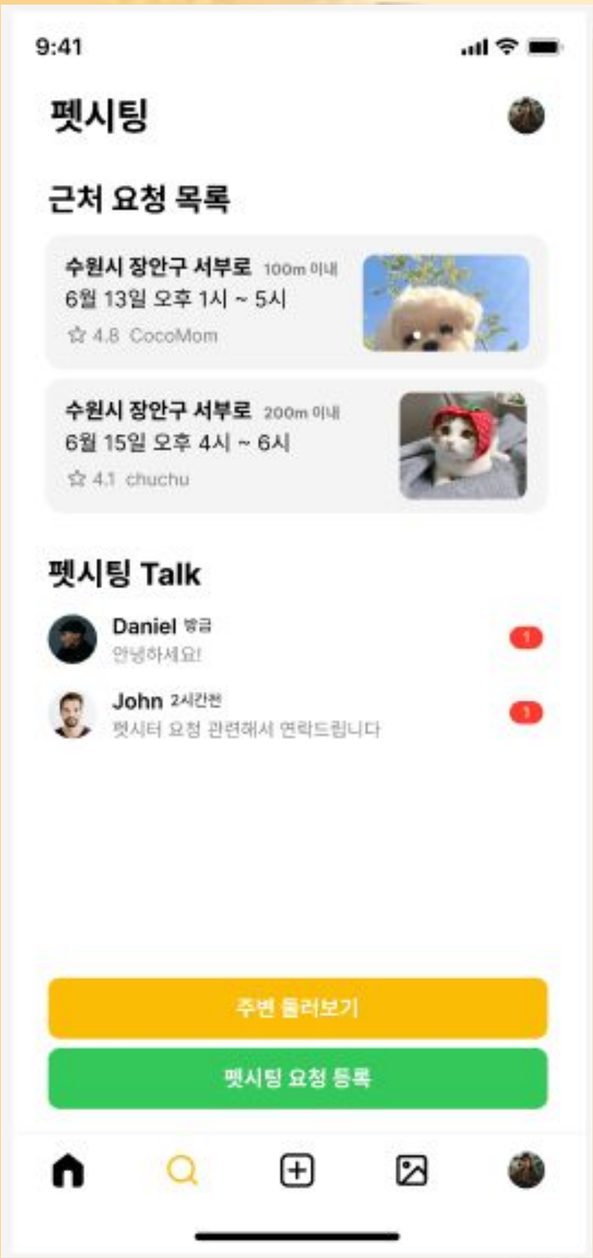
"He's at the side gate! Let me go there and give him some treats :)"



Scenario 2.

"I wonder if there is someone nearby who needs someone to look after their pet while they're gone?"

"I can make quick money while looking after cute pets! :)"



Scenario 3.

"I want to help the animals in the shelter!"

Scenario 4.

"I want to share photos of my cute cat to my friends!"

"I wish there was a space where we could share our pets and connect over the campus cats"

Demo Video

Expected Effect

01 Alleviate Loneliness

02 Promote Responsibility

03 Promote Mental Health

Thank You

For Listing